

Market Regime Detector + Portfolio Optimizer

An end-to-end AI system that detects stock market conditions and optimizes investment portfolios.

Phases: 1. Data Pipeline | 2. Feature Engineering | 3. Regime Detection | 4. Portfolio Optimization | 5. Backtesting | 6. Dashboard

Project Overview

This project uses machine learning to identify the "regime" of the stock market (Bull, Bear, or Sideways) and automatically suggests the best way to distribute your money across 10 different assets.

The Three Regimes

■ **Bull Market:** Prices rising, low volatility. Target: Growth.

■ **Bear Market:** Prices falling, high fear. Target: Capital preservation (Gold/Bonds).

↔ ■ **Sideways Market:** No clear direction. Target: Balanced diversification.

How It Works

Phase 1: Data Pipeline

Downloads 15+ years of daily prices and economic data (VIX, Inflation, Yields) and caches them for speed.

Phase 2: Feature Engineering

Creates 15 special signals (volatility, momentum, macro shifts) that describe the market's internal state.

Phase 3: Regime Detection (AI)

A Hidden Markov Model (HMM) analyzes the signals to classify each day as Bull, Bear, or Sideways.

Phase 4: Portfolio Optimization

Decides how much to invest in each asset to maximize Sharpe Ratio (Bull/Sideways) or minimize Volatility (Bear) between 5%-30% bounds.

Phase 5: Backtesting (Completed)

Accurate historical simulation spanning 15+ years using vectorbt with continuous rebalancing and friction modeling to accurately measure performance vs. Benchmark.

Technical Details

Feature Group	Metrics
Volatility	Daily returns, 20-day vs 60-day vol ratio
Momentum	1, 3, and 6-month trend strength
Macro	VIX fear index, Yield curve spread, Inflation
Cross-Asset	Bond-Equity correlation, Commodity trends

Status Update

Phases 1 through 5 are 100% complete and verified with 137 unit tests. We successfully proved via vectorbt that the dynamic combination of HMM Regime Detection and Portfolio Optimization delivers superior risk-adjusted returns (higher Sortino and Sharpe, lower Drawdown).

